

# ABRAR SHARIF MUDHIR

SENIOR SOFTWARE ENGINEER | PYTHON • JAVA • C# • AI/ML • FINANCIAL SERVICES

## PROFILE

GITHUB: [abrarmudhir.github.io](https://github.com/abrarmudhir)

LAST UPDATED: 04-Sep-2026

Senior Software and AI Engineer with 15+ years' experience delivering mission-critical systems across financial services and trading. Strong hands-on expertise in Python, Java, APIs, cloud-native platforms, and machine learning. Experienced technical leader in enterprise architecture, legacy modernisation, and engineering standards. Microsoft-certified AI Engineer and Data Scientist with a strong focus on Generative AI and AI-assisted development.

## EDUCATION

University College London, London, UK

SEP 2004 – JUN 2008

*Master of Engineering - M.Eng. (Hons) in Computer Science (2:1)*

- Emphasis on practical experience. Strong in software engineering, system design, and computational complexity.
- **Machine Learning Project:** *Investigation of correlated movement of exchange rates using neural networks.*

## CERTIFICATIONS

Microsoft, Online, UK

APR 2022 – PRESENT

*Associate*

### AI & DATA CERTIFICATIONS ([11 Active](#)):

- AI Engineer Associate ([AI-102](#)) • Data Scientist Associate ([DP-100](#))
- Data Engineer Associate ([DP-203](#)) • Database Administrator Associate ([DP-300](#))
- Power Platform Developer Associate ([PL-400](#)) + More

## EXPERIENCE

Brevan Howard Asset Management, London, UK

FEB 2026 – AUG 2026

*Quantitative Developer*

Designed and developed production Python applications, backend data services and analytics platforms supporting front-office investment and trading teams. Built scalable cloud-native services and data pipelines across AWS, Snowflake and MongoDB, translating complex business requirements into reliable, maintainable and supportable production solutions. Contributed across the full engineering lifecycle, including development, testing, CI/CD, deployment, monitoring and production support. Used AI-assisted engineering tools including Claude, Codex and Cursor to accelerate development, refactoring, automated testing and technical analysis while maintaining production engineering standards.

**SKILLS:** • Python Engineering (Python, SQL, backend services, APIs, production applications, data pipelines)

- Cloud & DevOps (AWS, Docker, CI/CD, deployment, monitoring, production support)
- Databases (Snowflake, MongoDB, NoSQL)
- Orchestration & Analytics (Prefect, Dash, batch and real-time data processing)
- AI-Assisted Engineering (Claude, Codex, Cursor, code generation, refactoring, testing)
- Engineering Practices (maintainability, reliability, data quality, troubleshooting, operational ownership)
- Stakeholders (portfolio managers, traders and analysts)
- Financial Markets (Fixed Income Credit, Convertible Bonds, MONIS, market and portfolio data)

**EDF Trading, London, UK**  
**Senior Data Engineer - CONTRACT**

**MAY 2025 – FEB 2026**

Designed and delivered production-grade Python and SQL data solutions across Gas, Power, and LNG trading, building scalable analytics pipelines and data products used by front-office teams. Worked closely with business stakeholders to translate trading requirements into reliable, maintainable technology solutions, with a strong focus on orchestration, data quality, and operational resilience.

**SKILLS:** • Data Engineering (Python, SQL, Spark, Hadoop, Polars, Pandas) • Cloud (Azure) • Pipelines & Orchestration (Dagster) • Databases (SQL Server) • BI (Tableau, Dash, Plotly) • Analytics Domains (Gas, Power & LNG Trading Analytics, Market Data, Time-Series Modelling) • Collaboration (Stakeholder Engagement, Mentoring, Technical Leadership).

**Pemberton Asset Management, London, UK**  
**Senior Python Developer - CONTRACT**

**SEP 2024 – MAY 2025**

Led the architecture and development of a production Python fund analytics platform, replacing approximately 20 spreadsheet-based models with a scalable, governed application. Designed reusable FastAPI REST APIs, backend services and data models using Python, SQLAlchemy and SQL Server, with Dash/Plotly interfaces. Developed strongly typed FastAPI services using Pydantic validation and SQLAlchemy, with Pytest automated testing. Worked directly with investment stakeholders to translate complex business rules into maintainable software, validate calculation outputs and improve the reliability, consistency and scalability of fund analytics.

**SKILLS:** • Domain (Private Credit, Fund Analytics, Carry, and Time-Series Modelling) • API Development (Python, FAST API) • User Interfaces (Dash, Plotly, HTMX) • Cloud (Azure) • Databases (SQL Server) • Collaboration (Stakeholder Engagement, Mentoring, Technical Leadership)

**Illio Technology Ltd, London, UK**  
**Senior Data Engineer**

**AUG 2023 – JUL 2024**

Led architecture and development of a cloud-native analytics and application platform, combining Python, APIs, AWS serverless technologies, microservices and data engineering.

**ACHIEVEMENTS:**

- Project: Data Integrations and APIs - Designed and developed Python/FastAPI microservices and REST APIs integrating external data providers, deploying services through Docker/Kubernetes and AWS Lambda/API Gateway.
- Technical Leadership & Team Development: Built the data function from the ground up, hiring and developing a team that included a mid-level data engineer and a data scientist. Provided technical leadership, coaching, and mentorship.
- Project: Serverless Data Pipelines - Implementing cloud-native ETL pipelines and a Data Analytics and Insights platform by leveraging key AWS services and Apache Airflow for Orchestration.
- Project: Analytics and Insights - Building an analytics and insights framework to facilitate faster time-to-market of Pythonic functions. Implementing analytics for Equities, FX, and Fund/ETF asset classes, focusing on financial markets and client multi-asset portfolios, including Cryptocurrencies.

**SKILLS:** • Data Engineering on AWS (AWS Glue (ETL, Crawlers, Catalog, Workflow), Apache Airflow, Python, Pandas, SQL Alchemy, Fast API, Lambda, PostgreSQL, Data Lake - S3) • Full Stack (Python, React, Kotlin, Kubernetes) • DevOps (AWS CDK, GitLab) • Cloud (AWS (EC2, SNS, SQS, RDS, Redshift))

Lead Engineer for a mission-critical investment modelling platform used by Fixed Income investment teams. Led engineers while remaining hands-on across Python application development, quantitative model implementation, platform architecture, APIs, and secure enterprise engineering.

**ACHIEVEMENTS:**

- Technical Leadership & Engineering Standards - Led and mentored three Senior Quant Developers, established engineering practices, supported architecture decisions, and coordinated delivery across multiple investment teams.
- Machine Learning / Quantitative Model Engineering - Partnered with Quant Researchers to productionise alpha, transaction-cost, risk, and optimisation models on a Python platform across Fixed Income, Equities, Macro and Credit; enhanced model back-testing and validation capabilities.
- Project: Aladdin Alpha Platform (Dynamic Flow) - Led development of reusable components within the Aladdin Alpha Platform and integrated Snowflake-based data capabilities to improve scalability and analytical workflows.
- Project: Defence in Depth - Led security modernisation across six mission-critical investment applications, redesigning interfaces and implementing secure APIs across Python, Java, .NET, MATLAB, and Perl while aligning solutions with enterprise security standards and controls.

**SKILLS:** • Leadership (Change Management, People Management, Project Management) • Practices (Agile, BDD, TDD) • Quant Stack (Python, Pandas, NumPy, Numba, Dask, Java, C#, MATLAB, SAS) • BI (Tableau, Microsoft Power BI) • Cloud (Microsoft Azure, Azure VM, Azure Functions) • Scheduling (Autosys)

Worked on a twelve-month contract with the European Space Agency (ESA), contributing to the Advanced Research in Telecommunications Systems Management Information System (ARTES MIS) project.

**ACHIEVEMENTS:**

- Project: ARTES MIS - Designed and developed Java/Spring Boot REST services and React/JSF applications supporting 350+ European Space Agency R&D activities, working within a secure private-cloud enterprise environment.
- BAU: Best Practices Rollout - Helped migrate applications into a common cloud environment, standardising development practices and improving maintainability, supportability, and cross-team knowledge sharing.

**SKILLS:** • Full Stack (Java, React, Spring Boot, Node.js, MariaDB) • Practices (Agile, BDD, TDD) • CMS (SharePoint) • BI (Microsoft Power BI) • Private Cloud (Private Network, Kubernetes, Docker, LDAP)

Worked as a Technical SME and Project Manager, handling budgeting, forecasting, stakeholder communication, and planning. Managed a team of five in migrating Golden Source to a cloud-hosted environment and led the upgrade of the legacy Security Master and Pricing product.

**ACHIEVEMENTS:**

- Project: M&G Prudential, Lite Security Master, and Pricing - Golden Source Upgrade - Delivered technical architecture advice and crafted functional and business requirements, technical solutions, and architecture documentation as a Technical SME and BA.
- BAU: Best Practices Rollout - Successfully advocated for and implemented DevOps practices and tools, enhancing Continuous Integration/Delivery (CI/CD), and streamlining delivery and regression testing.

**SKILLS:** • Leadership (Project Management, Microsoft Project) • Practices (Agile, JIRA) • GoldenSource EDM (Workflows, GSO/Mapping Designer, Connections) • Full Stack (Python, Java, Databases) • Integrations (Aladdin) • BI (Web Focus)

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**BlackRock, London, UK**

**NOV 2017 – MAR 2019**

**Associate, Senior Data Engineer of Aladdin Core Data**

Worked as a Senior Data Engineer, served internal investment teams, and fulfilled data product needs.

#### ACHIEVEMENTS:

- Project: Decommissioning of Data Management Platform (DMP) - Legacy Application Modernisation: Re-architected a mission-critical J2EE/Java monolithic data platform into scalable Python microservices and reusable data services, defining new application interfaces, data models, and integration patterns.
- Project: Large Datasets Pipelines - Increased data processing efficiency through the development of PySpark and Scala data pipelines, effectively managing large datasets on the Hadoop Distributed File System (HDFS).
- Project: Web Scrappers - Designed a reusable Java/Spring application framework for external data acquisition, enabling development teams to rapidly introduce new integrations using standardised components.
- Project: Decommissioning of MALT Refinery, Timeseries MIS - Modernised a legacy C# desktop application into a distributed Angular and Java Spring Boot platform, improving maintainability, accessibility, and extensibility.

**SKILLS:** • Data Engineering (Data Modelling, Data Lake, Hadoop, Apache Spark) • Full Stack (Python, Java, C#, Spring Boot) • Practices (Agile, BDD, TDD) • DevOps (GIT, Sonar) • Scheduling (Autosys)

**Golden Source Limited, London, UK**

**JUL 2016 – NOV 2017**

**Senior Technical Consultant, Professional Services**

Worked as a senior technical consultant and SME and led GoldenSource implementations for clients. Led a 4-person team to validate intraday pricing rates, curves, and surfaces in a Reference and Market Data Project. Led a 4-month Fundamental Review of the Trading Book (FRTB) Proof of Concept (POC).

#### ACHIEVEMENTS:

- Project: Siemens Financial Services, Market Data Solution Implementation - Built a reusable Murex connector and R analytics library with potential for sale to other clients. Utilized WebFOCUS (GS Insight) for BI reports, enhancing data accuracy and reporting efficiency.
- Project: ICBC Standard Bank, Fundamental Review of Trading PoC - Integrated C-based scenario-generation libraries into the GoldenSource platform to support stress testing and persist outputs within the Risk Master Data Model, contributing to the delivery of an FRTB-aligned risk analytics solution.

**SKILLS:** • Full Stack (Python, JEE, PSSQL (Oracle), R) • Golden Source EDM (Orchestrator, Workflows, Mapping Designer, Market Data Solution)

**Rapid Addition Limited, London, UK**

**OCT 2014 – JUL 2016**

**Senior Developer, Trading Technology**

Developed mission-critical low-latency trading infrastructure in Java and C#, implementing core functionality within commercial FIX/FAST engine libraries used for electronic market connectivity. Contributed a new full-stack ASIC reporting product, utilizing the FIX protocol. Extended RA ShortCut – FIX for BizTalk, enhancing performance and resolving bugs.

#### ACHIEVEMENTS:

- Project: RA FIX Engine - Improved API functionality, code quality and test coverage through engineering reviews and automated testing. Developed trading solutions, FIX, and FAST engines, streamlining processes and enhancing stakeholder satisfaction.
- Project: RA ASIC Short Reporting Tool - Implemented a Full-stack solution for submitting short-selling reports to ASIC, enhancing efficiency through optimized FIX protocol utilization. Utilized ASP.NET and C# for the front end and MSSQL for the back end, ensuring seamless deployment on Azure Cloud, and reducing deployment time.

**SKILLS:** • Full Stack (Java, .NET (C#, ASP.NET)) • FIX (Protocol/Engine, Certification)  
• Middleware (Microsoft BizTalk) • Cloud (Microsoft Azure)

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**BlackRock, London, UK**

**AUG 2010 – JUL 2014**

**Associate, Senior Quant Developer of Model-Based Fixed Income**

Worked as a quant developer, implemented fixed-income investment models, optimizing efficiency, ensuring accurate data feeds, and managing the entire model lifecycle from development to production.

#### ACHIEVEMENTS:

- Project: Common Alpha-Gen (CAGE) Platform- Developed a reusable quantitative execution framework supporting models written in SAS, R and MATLAB and integration with enterprise investment systems.
- Project: Machine Learning & Quantitative Model Productionisation - Collaborated with quantitative researchers to implement, back-test and productionise alpha, risk, and transaction-cost models within a Fixed Income investment platform, managing the model lifecycle from research through production.
- Project: Data Management Platform- Engineered Data Management tools for efficient slicing and dicing of input data, delivering Tableau reports for front-office portfolio managers to enhance monthly attribution and investment dataset views.

**SKILLS:** • Data Engineering (Informatica, J2SE/JEE, JavaScript) • Quant Stack (MATLAB, SAS, R, C#) • BI (PowerBI, Tableau, SAP Business Objects) • Scheduling (Autosys)

**Client Knowledge Limited (now [ICAP](#)), London, UK**

**MAY 2009 – AUG 2010**

**Quantitative Developer, Managed Models**

Worked as a quant developer and connectivity engineer, facilitating real-time market connectivity and high-frequency algorithmic trading.

#### ACHIEVEMENTS:

- Project: Market Connectors - Designed and developed a low-latency framework to capture, receive, and replay real-time high-frequency data, reducing reliance on third-party market data systems. Developed adapters for CEP Engines enabling direct access to streaming market data using FIX technologies. Connected to various forex market data feeds and the Market News Dow Jones feed.
- Project: Model Implementation - Collaborated with research analysts to implement quantitative models, including Value at Risk (VaR) and FX trading models in CEP engines (Progress Apama and Coral8).

**SKILLS:** • Full Stack (Java, JavaScript, Progress Apama CEP) • FIX (QuickFix, Client Onboarding, Testing) • Data Engineering (Data Modelling, SQL) • BI (SAP Business Objects BI 4. X).

**Barclays Global Investors (now [BlackRock](#)), London, UK**

**JUN 2008 – SEP 2008**

**SOA Performance Analyst, Enterprise Architecture - INTERNSHIP**

**SKILLS:** • Full Stack (Java, Progress Sonic Workbench, JMS, Streaming, JBoss/Tomcat containers)

**British Telecom, Adastral Park, Ipswich, UK**

**JUL 2007 – SEP 2007**

**Rich Client User Interface Designer, Innovation Lab - INTERNSHIP**

**SKILLS:** • Full Stack (Java EE, JSPs, SIP Servlets, XML, HTML, CSS, JavaScript, AJAX, Web 2.0)

**Sempra Energy, London, UK**

**AUG 2006 – OCT 2006**

**Financial Application Developer, Energy Trading - INTERNSHIP**

**SKILLS:** • Full Stack (Excel VBA, Java, XML processing, HTML, CSS, JavaScript)